

Lab 4: LLM Chat History

Connect Four vs AI - coding-assistant record | 30 September 2026

Repository	github.com/2306apk/15_connect_four
Scope	Win detection, termination, AI decisions, move feedback, tests, and submission artifacts
Result	15 automated tests passed

Complete conversation link

https://chatgpt.com/s/cx_6abd3ec9fe4c8191b879502bdf8cdc16

The public link above is the canonical complete chat record. This PDF provides a compact export of the request, audit trail, implementation decisions, and verification results.

User request

"Refer to the PDF in the file, complete all deliverables, and push to my repository."

Assistant audit and actions

- **Inspected first:** read the handout, README, all Python modules, repository history, and baseline runtime behavior before editing.
- **Reproduced defects:** confirmed diagonal wins were missed and the AI selected random legal columns without win-or-block reasoning.
- **Implemented:** added both diagonal directions, safe simulated AI moves, immediate win priority, immediate block priority, draw handling, and exactly one feedback line per successful move.
- **Preserved architecture:** modified the existing Board, AI, and Game modules; added no runtime dependency or persistent storage.
- **Added tests:** covered horizontal, vertical, both diagonals, shorter sequences, full and invalid columns, draw state, invalid input, quitting, AI wins, AI blocks, legal choices, feedback, and immediate termination.
- **Prepared submission:** generated before and after 10-second MP4 demonstrations and this chat-history export.

Verification Record

Automated checks executed against the final code

Area	Cases	Status
Win detection	Horizontal, vertical, down-right, down-left, length-three negative case	PASS
Board safety	Out-of-range column, full column, board-full state	PASS
AI behavior	Immediate win, immediate block, only legal move, no legal move	PASS
Game flow	Invalid input, quit, one feedback per move, stop immediately after win	PASS
Static checks	Python compilation and git diff whitespace check	PASS
Videos	before.mp4 and after.mp4: H.264, 1280x720, 10.0 seconds each	PASS

Commands used for final verification

```
python3 -m unittest discover -s tests -v
```

```
python3 -m py_compile main.py game.py board.py ai.py tests/test_connect_four.py
```

```
git diff --check
```

```
ffprobe -show_entries format=duration,size Lab-4/Videos/before.mp4
```

```
ffprobe -show_entries format=duration,size Lab-4/Videos/after.mp4
```

Submission note

The repository contains the updated source, test suite, two required videos, this PDF export, and a Markdown copy of the public chat link under the Lab-4 folder.